Homework 21

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1 $Premises : q \implies r, (r \vee k) \implies (m \vee m).Prove : q \implies m$

1.1 Prove by Indirect deduction

$$
\begin{aligned}
&q \implies r && Premise && (1)\\
&(r \vee k) \implies (m \vee m) && Premise && (2)\\
&\neg(q \implies m) && Assumtion && (3)\\
&\neg(\neg q \vee m) && 3\ MaterialImplication && (4)\\
&q \wedge \neg m && 4\ DeMorgan, DoubleNegation && (5)\\
&\neg m && 5\ Simplication && (6)\\
&(r \vee k) \implies m && 2\ Idempotent && (7)\\
&\neg(r \vee k) && 6, 7\ ModusTollens && (8)\\
&\neg r \wedge \neg k && 8\ DeMorgan && (9)\\
&\neg r && 9\ Simplication && (10)\\
&q && 5\ Simplication && (11)\\
&\neg q && 1, 10\ ModusTollens && (12)\\
&False && 12, 13\ Identity && (13)
\end{aligned}
$$

$\therefore q \implies m$